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Invoiveocr: deep learning based automatic invoice recognition process

Chao Wang[1*], Jian Zhang[1] and Yu Yu[1]

*Correspondence:
Chao Wang
wangchaosgcc@qq.com
[1]State Grid Information & Telecommunication Group, Beijing Guodiantong Network Technology Co., Ltd., Beijing, China

Abstract

In recent years, the automatic recognition of invoices has emerged as a critical component in document digitization, particularly within the financial sector, where efficient data extraction from structured documents is essential. This paper introduces an advanced method for automating invoice recognition by combining cutting-edge image preprocessing, OCR pipeline techniques, and post-processing strategies. The process begins with a robust image preprocessing stage that includes scaling, binarization, noise reduction, and skew correction based on Hough transform; learning-based angle regressors may offer adaptive alternatives for noisy or multi-aligned invoices, all of which enhance the quality of the input images, ensuring optimal performance during OCR. The core of the system is an OCR pipeline that integrates state-of-the-art text detection and recognition algorithms, effectively handling various challenges such as multi-oriented and densely packed text regions. After the initial recognition, a series of post-processing steps, including error correction, field validation, and structured data formatting, are applied to ensure that the extracted text is accurate and conforms to predefined invoice formats. Extensive experiments conducted on a dataset comprising 6349 images, including both electronic and paper invoices, demonstrate that the proposed approach achieves high recognition accuracy and reliable performance. These results highlight the system's effectiveness in automating invoice recognition, confirming its applicability in real-world financial applications where precision and efficiency are paramount. The method provides a comprehensive solution for improving invoice digitization and streamlining financial workflows.

Keywords Optical character recognition, Text detection algorithms, Text recognition algorithms, Structured data extraction, Invoice recognition

1 Introduction

With the acceleration of digital transformation, industries across the board are actively exploring the adoption of advanced information technologies to enhance work efficiency and management standards [1]. In the electric power industry, particularly in the area of bill management for power grid enterprises, traditional manual identification methods are becoming increasingly inadequate due to their inefficiency and susceptibility to errors. As a key foundation for power trading [2], settlement, and financial management,

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the accuracy and efficiency of bill processing directly impacts operational costs and customer satisfaction.

In recent years, Optical Character Recognition (OCR) technology has gained widespread use across various sectors, including finance, insurance, healthcare, and transportation. OCR technology [3] significantly streamlines the process of text data entry by converting text from paper documents or images into editable formats. However, the application of OCR in bill management within power grid enterprises remains in its early stages, particularly when it comes to recognizing bills in complex environments, such as poor lighting, suboptimal shooting angles, and low-quality paper. In these cases, recognition accuracy needs further improvement [4].

Studies have shown that the accuracy and efficiency of text recognition can be greatly enhanced by leveraging pre-trained OCR large models [5]. Research suggests that deep learning-based OCR techniques hold great potential in text recognition [6], particularly when applied to large and diverse datasets. However, most of these studies focus on general-purpose text recognition, with relatively few addressing the specific needs of power grid enterprises.

In recent years, advancements in Optical Character Recognition (OCR) technologies have significantly improved the accuracy and efficiency of text detection and recognition in complex document formats such as electronic invoices [7]. Traditional OCR systems often struggled with challenges posed by variable text orientations, dense layouts, and irregular text shapes. However, modern algorithms such as EAST (Efficient and Accurate Scene Text Detector), DB (Differentiable Binarization), and SAST (Segmentation-based Scene Text Detector) have revolutionized the text detection phase by offering robust solutions for handling multi-oriented, curved, and densely packed text regions. These detection methods, when combined with powerful recognition models like CRNN (Convolutional Recurrent Neural Network), which captures sequential text information, and RARE (Robust Attention-based Recognition), which excels in recognizing distorted or irregular text, enable highly accurate and efficient OCR pipelines. By integrating these algorithms, the recognition of complex document structures—such as invoices—has become more reliable, ensuring that extracted data is both precise and well-structured for downstream applications. This progress has laid the foundation for more effective document digitization and automation in financial systems, driving improvements in areas such as invoice processing and data extraction.

Against this backdrop, this study seeks to address the inefficiencies and error-prone nature of manual bill recognition in power grid enterprises by introducing an intelligent bill recognition and verification method based on a pre-trained OCR macromodel. This study [8] follows a systematic approach: first, by collecting and annotating large-scale image-text datasets; second, by conducting model training and optimization; and finally, by validating the model's recognition performance through experimental testing.

This research not only provides technical support for the refined management of smart grids but also promotes the automation and standardization of bill management processes [9]. Specifically, by enhancing the accuracy and efficiency of bill recognition, this study significantly improves business processing efficiency, reduces labor costs, and minimizes human error, thereby providing strong support for the refined management of smart grids. Furthermore, this study is expected to encourage broader

applications of OCR technology in other fields, facilitating the deeper integration of digital transformation.

In conclusion, this study not only expands the application scenarios of OCR technology by conducting in-depth research into its use in bill management for power grid enterprises but also offers practical solutions for improving operational efficiency and management standards within these enterprises. This has significant theoretical and practical implications.

2 Methodology

This section introduces our proposed automatic electronic invoice recognition model based on the OCR foundational model—InvoiceOCR. The structure of the model is shown in Fig. 1. In the OCR recognition system for electronic invoices, the process begins with image preprocessing, where techniques such as image scaling, binarization, noise removal, and skew correction based on Hough transform; learning-based angle regressors may offer adaptive alternatives for noisy or multi-aligned invoices are applied to optimize the input to meet the assumptions of downstream models like CRNN and SRN, which require horizontally aligned, grayscale inputs for stable decoding. This is followed by the OCR pipeline, which includes text detection using algorithms like EAST [10], DB [11], and SAST [12] to identify text regions, and text recognition models such as CRNN [13], RARE [14], and SRN [15] to accurately extract textual information from the detected regions. Finally, the post-processing module refines the recognized text through error correction, field validation, structured data formatting, and entity

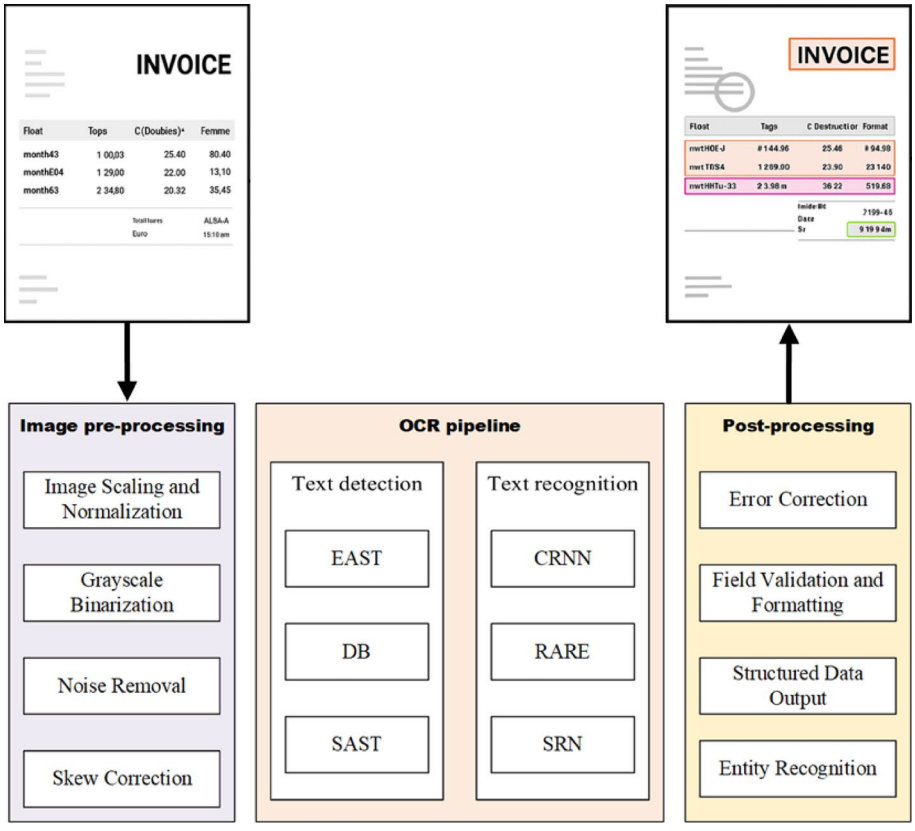


Fig. 1 Illustration of the InvoiceOCR pipeline architecture including preprocessing, detection, recognition, and post-processing modules

recognition, ensuring that the output is accurate, well-organized, and ready for integration into downstream applications.

2.1 Image pre-processing module

In the image preprocessing stage of electronic invoice recognition, several essential techniques are applied to optimize the input for the OCR model. First, the images are subjected to scaling and normalization [16], where each image is resized to a standardized dimension using bilinear interpolation, and padded using zero-padding to preserve content at borders to ensure consistency across the dataset, eliminating size variability and improving downstream processing.

$$I_{\text{resized}} = resize\left(I_{input}, (h, w)\right) \tag{1}$$

$$I_{norm} = \frac{I_{resized}}{255} \tag{2}$$

This formula resizes the input image I_{input} to the target dimensions (h, w). Here, h represents the target height, and w represents the target width.

Next, the images undergo grayscale conversion and adaptive binarization techniques such as Otsu or Sauvola methods, which adjust thresholds based on local contrast [17] to enhance the contrast between text and background, making the textual information more prominent and easier for the OCR engine to detect.

$$I_{gray} = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \tag{3}$$

This normalizes the pixel values to the range [0,1]. I_{resized} is the resized image, and 255 is the maximum possible pixel value.

$$I_{\text{binary}}\left(x, y\right) = \begin{cases} 255, if I_{gray}\left(x, y\right) > T \\ 0, if I_{gray}\left(x, y\right) \leq T \end{cases} \tag{4}$$

This binarizes the grayscale image by applying a threshold T. If the grayscale pixel value $I_{gray}\left(x, y\right)$ is greater than T, it is set to 255 (white); otherwise, it is set to 0 (black).

Following this, noise removal techniques, such as Gaussian filtering with empirically chosen kernel size (e.g., 3×3), balancing noise suppression and text clarity [18], are employed to eliminate unnecessary artifacts that may obscure the text, improving the clarity of the image.

$$I_{filtered}\left(x, y\right) = \sum_{i=-k}^{k} \sum_{j=-k}^{k} I_{binary}\left(x + i, y + j\right) \cdot G\left(i, j\right) \tag{5}$$

This applies a Gaussian filter to the binary image for noise removal. $I_{binary}\left(x + i, y + j\right)$ is the value of the binary image at position $(x + i, y + j)$, and $G\left(i, j\right)$ is the Gaussian kernel. The parameter k defines the kernel size.

Lastly, skew correction based on Hough transform; learning-based angle regressors may offer adaptive alternatives for noisy or multi-aligned invoices [19] is applied to adjust any tilted or misaligned text lines, ensuring that the text appears horizontally aligned. For multi-angle zones, rotation-invariant or affine normalization techniques could further improve robustness, which enhances the accuracy of the OCR recognition

process. These preprocessing steps are critical for enhancing image quality and improving the overall performance of the OCR system.

$$I_{corrected} = rotate\left(I_{filtered}, -\theta\right) \tag{6}$$

This formula corrects skew in the image by rotating it by an angle $-\theta$. Here, θ is the estimated skew angle, and $I_{filtered}$ is the filtered image to be rotated.

2.2 OCR pipeline

In the OCR pipeline for electronic invoice recognition, two primary stages are involved: text detection and text recognition. The text detection phase employs several advanced algorithms, including EAST, DB, and SAST, to accurately identify the regions in the image that contain text. EAST (Efficient and Accurate Scene Text) [10] is a robust and fast algorithm used for detecting text regions in arbitrary orientations, while DB (Differentiable Binarization) [11] and SAST (Segmentation-based Scene Text) [12] further refine the detection process, offering enhanced precision, particularly in dense or curved text regions. Specifically, EAST is a robust and fast algorithm designed for detecting text regions in arbitrary orientations. Unlike traditional methods that involve multiple processing steps, EAST directly predicts text regions and bounding boxes using a fully convolutional neural network (FCN). This simplicity results in faster inference times while maintaining high accuracy. EAST is an anchor-free detector, particularly suited for scenarios with irregular text layouts, making it ideal for detecting text in complex documents such as electronic invoices. EAST algorithm uses a multi-task loss, where geometry and score losses are normalized to prevent imbalance during training that combines geometry and segmentation losses, typically expressed as:

$$L_{EAST} = L_{geometry} + \lambda \cdot L_{score} \tag{7}$$

$L_{geometry}$ is the geometry loss for predicting the shape of the text region, and L_{score} is the score loss for measuring the confidence of the detected text region. λ is a weight parameter.

Differentiable Binarization (DB) is a state-of-the-art algorithm that refines the text detection process by introducing a differentiable binarization module. This module helps separate text from the background more effectively, especially in dense and curved text regions. DB simplifies the detection pipeline and improves accuracy by leveraging a binary map that differentiates between text and non-text regions, making it highly effective in handling dense text layouts found in electronic invoices. DB optimizes text detection through a binarization loss, represented as:

$$L_{DB} = L_{\text{binary}} + \alpha \cdot L_{threshold} + \beta \cdot L_{similarity} \tag{8}$$

L_{binary} is the binarization loss, $L_{threshold}$ is the threshold loss, and $L_{similarity}$ measures the similarity between predicted and ground truth regions. α and β are weighting factors for the respective terms.

SAST is a segmentation-based text detection algorithm designed to handle irregular and complex text shapes. By treating text detection as a segmentation problem, SAST can accurately detect curved and multi-oriented text regions, which are often challenging for traditional detection models. This makes SAST well-suited for recognizing text in complex documents like electronic invoices, where text may appear in varying

orientations and formats. The SAST algorithm uses a segmentation-based loss function to handle complex text detection, expressed as:

$$L_{SAST} = L_{segmentation} + \gamma \cdot L_{orientation} \tag{9}$$

$L_{segmentation}$ is the segmentation loss for differentiating text from the background, and $L_{orientation}$ is the orientation loss for handling text in different directions. γ is a weighting parameter.

Following text detection, the identified regions are processed in the text recognition phase, where models like CRNN, RARE, and SRN are utilized to convert the detected text images into digital text. CRNN (Convolutional Recurrent Neural Network) [13] combines convolutional layers with recurrent networks to capture the sequential nature of text. CRNN combines convolutional and recurrent neural networks to perform text recognition tasks. It extracts spatial features from text images using convolutional layers and captures the sequential nature of text using recurrent layers. CRNN is highly effective for recognizing text in images with varying lengths and orientations, such as electronic invoices. It also utilizes Connectionist Temporal Classification (CTC) loss to handle variable-length text sequences without requiring strict alignment between input and output. The CTC loss is represented as:

$$L_{CRNN} = -\log P\left(Y|X\right) \tag{10}$$

$P\left(Y|X\right)$ is the conditional probability of the predicted text Y given the input image X.

RARE (Robust Attention-based Recognition) [14] focuses on complex, irregular text, using attention mechanisms to improve recognition accuracy. RARE is a powerful text recognition model that incorporates spatial transformers and attention mechanisms to handle irregular or distorted text. It dynamically adjusts the input text's geometry using a spatial transformer network, enabling the recognition of text with varying orientations and distortions. The attention mechanism further enhances recognition accuracy by focusing on the most relevant features in the text sequence. RARE is particularly effective for handling complex, non-linear text in electronic invoices. RARE uses an attention-based loss to improve the recognition of complex and irregular text, represented as:

$$L_{RARE} = L_{CTC} + \delta \cdot L_{attention} \tag{11}$$

L_{CTC} is the CTC loss, and $L_{attention}$ is the attention loss for improving recognition accuracy of distorted or complex text. δ is a weighting factor.

Finally, SRN (Sequence Recognition Network) [15] leverages advanced sequence-to-sequence learning for high-performance recognition, particularly in dealing with diverse font styles and text orientations. This pipeline ensures high-accuracy text recognition in complex document images like electronic invoices. SRN leverages sequence-to-sequence learning, using the following loss function:

$$L_{SRN} = L_{sequence-to-sequence} + \eta \cdot L_{alignment} \tag{12}$$

$L_{sequence-to-sequence}$ is the sequence-to-sequence loss, and $L_{alignment}$ is the alignment loss for handling text with varying fonts and orientations. η is a weight for the alignment term.

2.3 Post-processing module

In the post-processing phase of the OCR pipeline for electronic invoice recognition, several crucial steps are implemented to enhance the accuracy and structure of the extracted text data. The first step is error correction [20], which identifies and fixes common recognition errors, such as incorrect characters or symbols, ensuring that the extracted data aligns with expected patterns. The formulation is represented as:

$$Corrected_{text} = ErrorCorrection\,(x) \tag{13}$$

x represents the recognized text, and $ErrorCorrection\,(x)$ is the process that corrects common recognition errors, such as character misidentifications.

Following this, field validation and formatting ensure that the recognized text matches specific formats (e.g., date, invoice number, total amounts) and adheres to the predefined structure of an invoice. The formulation is represented as:

$$Validated_{field} = Validate\,(f_{recognized}, format) \tag{14}$$

$f_{recognized}$ is the recognized field (e.g., date, invoice number), and $format$ is the expected format. The function $Validate$ ensures that the recognized field matches the expected format.

Next, the recognized data is organized into a structured format in the structured data output step. This step ensures that the extracted information, such as invoice details, is stored in a machine-readable format (e.g., JSON, CSV) for easy integration with other systems. The formulation is represented as:

$$Structured_{output} = Organize\,(x_{recognized}) \tag{15}$$

$x_{recognized}$ refers to the recognized data, and $Organize$ is a function that converts this data into a structured format such as JSON or CSV.

Finally, entity recognition is employed to identify and label specific fields within the invoice, such as supplier name, invoice number, and tax information, providing context and meaning to the recognized text. The formulation is represented as:

$$Entities = RecognizeEntities\,(x_{recognized}) \tag{16}$$

$x_{recognized}$ is the recognized text, and $RecognizeEntities$ identifies and labels specific fields such as supplier name, invoice number, and tax information.

Together, these steps optimize the output of the OCR pipeline, ensuring that the extracted data is both accurate and useful for further processing.

3 Experimental results

3.1 Datasets description

To train a high-performance OCR model, it is essential to prepare large-scale and diverse image-text datasets. Thousands of paper and electronic invoices were collected from NEC to construct the dataset. OCR technology was utilized to extract text from the images, leveraging either open-source OCR libraries or cloud-based OCR services. Once the text was extracted, the image and corresponding text were annotated. The annotation process involved eit her directly linking the extracted text with the image or highlighting the text area within the image by drawing bounding boxes around the text

and associating the corresponding content. This approach allows the model to learn the relationships between image regions and the text they contain.

Following the data labeling, the dataset underwent preprocessing to enhance the model's performance during training. The data was cleaned to eliminate inaccurately or incompletely labeled samples. Automated tools were employed to facilitate this process, including the removal of duplicate data and correction of erroneous annotations. These steps ensure that the dataset is high-quality and optimized for training the OCR model.

In this study, a total of 6349 images were used as the dataset, consisting of 4618 electronic invoices and the remaining images representing paper invoices. The dataset was divided into a training set, validation set, and test set, following the commonly used ratio of 7:2:1. This division ensures that the model is trained on a substantial portion of the data, while also allowing for validation and testing to evaluate its performance and generalization ability on unseen data.

3.2 The performance of text detection

We conducted experiments to evaluate the performance of text detection in the OCR pipeline after image preprocessing, as shown in Fig. 2. The image on the top represents an electronic invoice, while the bottom side illustrates the OCR text detection results, where the text regions are highlighted and segmented. The OCR system successfully identifies and extracts several key fields from the invoice, including the invoice number, date, and various financial details. The text areas are precisely boxed, showcasing the system's capability to detect text in both horizontal and vertical orientations. Most of the text fields, such as invoice codes and monetary amounts, are captured accurately, although minor misalignments are observed in densely populated areas. Overall, the OCR detection is effective in extracting a comprehensive set of invoice details, which can subsequently be used in further stages of the pipeline, such as text recognition and data validation.

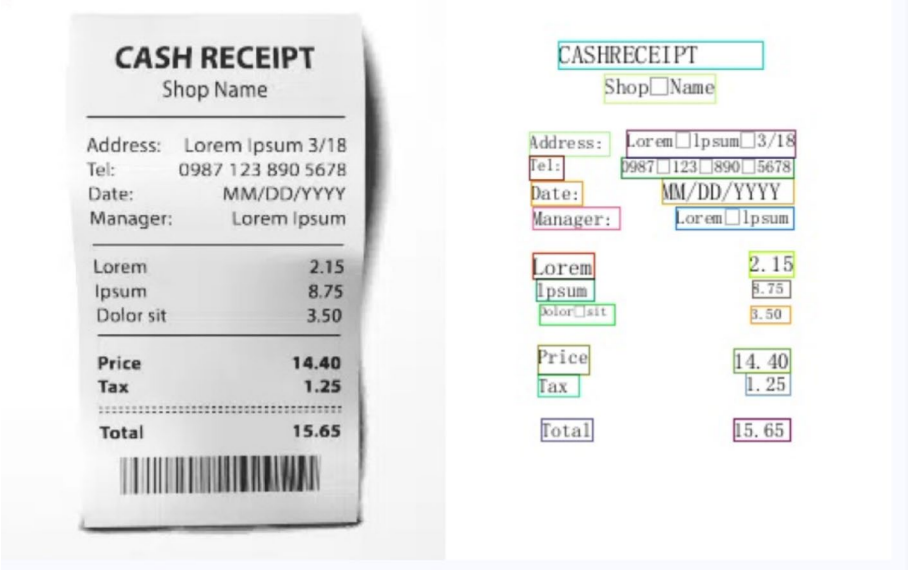


Fig. 2 The qualitative results of text detection

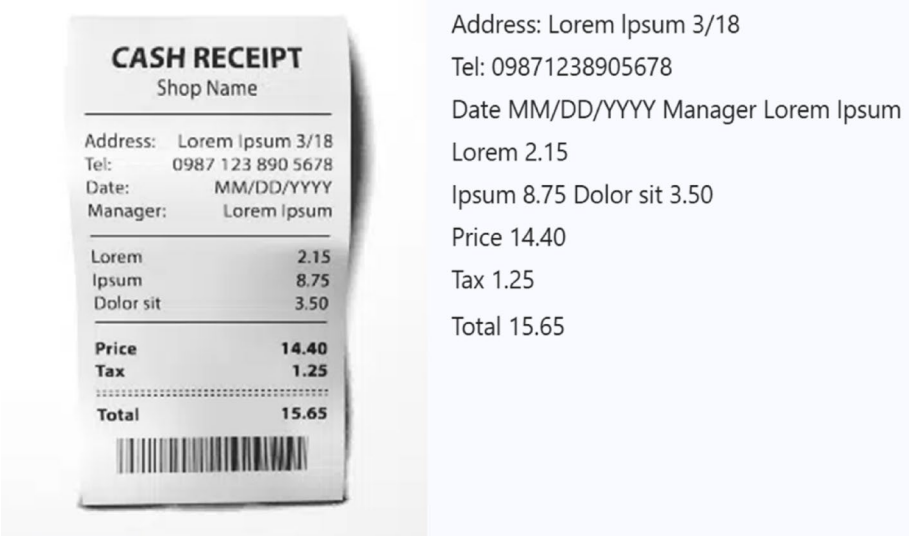


Fig. 3 The qualitative results of text recognition

Table 1 OCR recognizes paper invoices

Key	Value
Address	Lorem Ipsum
Tel	09871238905678
Lorem	2.15
Price	14.40
Tax	1.25
Total	15.65

3.3 The performance of text recognition

In the text recognition stage following text detection, we present the results in Fig. 3. The image represents a sample of text extracted from an invoice using OCR technology. The system accurately identified key elements such as the buyer's and seller's names, tax identification numbers, item descriptions, quantities, unit prices, and total amounts. Numeric values related to tax rates, totals, and prices were correctly captured, along with specific details like product names and technical specifications (e.g., wire sizes and equipment). However, some instances of overly precise decimal points in the numeric values suggest potential OCR over-precision, which might require post-processing to refine the results and eliminate unnecessary detail. Overall, the text recognition effectively captured essential information, though further improvements could enhance the clarity and usability of the extracted data, especially for numerical fields.

3.4 The performance of text output

Despite achieving relatively high accuracy in text detection and recognition after the image preprocessing module of InvoiceOCR, the final recognition results are only transformed into a series of corresponding dictionary files for querying and practical use after passing through the post-processing module. The keys and their corresponding sample values used in this study are shown in Table 1. It is evident that InvoiceOCR effectively extracts the desired keys and their corresponding values, demonstrating the effectiveness of automatic recognition technology for electronic invoices.

Table 2 The text recognition accuracy of paper and electronic tickets

Dataset	EAST	DB	SAST	Average
Paper ticket dataset	0.79	0.82	0.85	0.82
Electronic ticket dataset	0.92	0.98	0.98	0.96

Table 3 The field recognition accuracy of paper and electronic tickets

Dataset	Fields	CRNN	RARE	SRN	Average
Paper ticket dataset	Amount field	0.80	0.84	0.87	0.84
	other key fields	0.79	0.80	0.83	0.81
Electronic ticket dataset	Amount field	0.94	0.98	0.98	0.97
	other key fields	0.91	0.93	0.98	0.94

Table 4 The ablation study of paper and electronic tickets

Dataset	Baseline	+ Image scaling and normalization	+ Grayscale binarization	+ Noise removal	+ Skew correction
Paper ticket dataset	0.62	0.68	0.75	0.79	0.82
Electronic ticket dataset	0.74	0.82	0.88	0.93	0.96

Additionally, we conducted ablation quantitative experiments, as shown in Table 2, to validate the accuracy of the text recognition. Our proposed method achieved an accuracy of 0.82 on the Paper Ticket dataset and 0.96 on the Electronic Ticket dataset, further demonstrating the effectiveness of the approach. These results confirm the robustness and accuracy of our method in different invoice types, proving its applicability in real-world scenarios.

Text field recognition accuracy is one of the most important indicators for evaluating the performance of OCR technology. Table 3 provides a detailed description of the OCR text field recognition accuracy. The results highlight the effectiveness of our approach in accurately recognizing various text fields across different datasets. High accuracy in recognizing critical fields such as invoice numbers, dates, and amounts further demonstrates the reliability of our OCR system in practical applications, especially in handling structured documents like electronic and paper invoices.

As shown in Table 4, we conducted a series of ablation experiments to evaluate the impact of the preprocessing components on text recognition performance. As the preprocessing pipeline is progressively enhanced, the recognition accuracy consistently improves across both datasets. This demonstrates the effectiveness of each individual component and highlights the critical role of high-quality preprocessing in OCR tasks.

4 Discussion and conclusions

In this study, we explored the application of OCR-based intelligent ticket recognition and verification technology in the context of power grid ticket management. Our system is built upon a robust OCR pipeline that integrates advanced algorithms such as EAST for text detection and CRNN for text recognition, ensuring high precision in detecting and recognizing key fields within invoices. Through comprehensive experiments and case studies, the large OCR model demonstrated outstanding performance, achieving a recognition accuracy of 0.96 for electronic invoices and over 0.97 for key fields such as invoice numbers and dates. These results affirm the efficiency and accuracy of OCR technology, especially when using modern techniques like Differentiable Binarization

(DB) to improve text detection in dense or irregular regions, and attention-based models like RARE to handle more complex and distorted text sequences. This pipeline significantly enhances the automation of bill processing, far surpassing traditional manual methods that are prone to errors and inefficiencies.

Our proposed system is structured around a well-defined encoder-decoder architecture, where each module performs a specialized role to facilitate seamless interaction, supporting the smart grid's fine-grained management. The system's stability and reliability have been validated in complex business scenarios, where optimization techniques such as data augmentation, model selection, and hyperparameter tuning were applied to enhance performance further. These improvements not only increased accuracy but also reduced error rates, demonstrating the potential of OCR technology to revolutionize bill processing in the power grid domain.

However, limitations still exist, particularly regarding the system's recognition accuracy in complex environments, such as under poor lighting, suboptimal shooting angles, or degraded paper quality. Moreover, while the dataset used in this research is diverse, it primarily consists of invoices sourced from power grid suppliers, which may not fully represent the variety of bills encountered in broader real-world applications. Additionally, although this study focuses on the technical capabilities of OCR, important issues such as security and compliance—critical in grid ticket management—have been relatively underexplored.

In future work, efforts should focus on improving recognition accuracy in challenging environmental conditions by refining algorithms such as SAST, which could offer better segmentation of irregularly shaped text. Expanding the dataset to include a broader variety of bill types will also be essential for making the system more generalizable. Furthermore, addressing security and privacy concerns will be crucial to ensuring compliance with industry regulations. OCR technology has the potential to extend its application beyond power grid management into sectors like finance, insurance, healthcare, and education, paving the way for a more extensive digital transformation across industries. Additionally, incorporating transformer-based detection architectures could enhance multi-scale feature extraction. Exploring synthetic data augmentation using diffusion models may also improve robustness under low-sample conditions.

Author contribution
C.W. and J.Z. wrote the main manuscript text and Y.Y. prepared figures and table. All authors reviewed the manuscript.

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Declarations

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